# Particle Physics Homework 4

## 1). Complex scalar fields

Consider a complex scalar field $\phi = \frac{\phi_1 + i\phi_2}{\sqrt{2}}$ with a Lagrangian $\mathcal{L} = \partial_\mu \phi^* \partial^\mu \phi - m^2 \phi^* \phi$

1.1) Find the equations of motion for $\phi$ & $\phi^*$

1.2) In the case that $m = 0$, consider the field translation:

$$\phi(x^\mu) \rightarrow \phi'(x^\mu) = \phi(x^\mu) + \alpha$$

where $\alpha$ = constant for any point in the spacetime.

Show that this translation is a symmetry.

1.3) From 1.2), find the Noether current & conserved charge correspond to the symmetry.

## 2). Gauge symmetry with gauge fixing term

Consider EM Lagrangian with a term

$$\mathcal{L} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} - \frac{1}{2\alpha} \partial_\mu A^\mu \partial_\nu A^\nu ,$$

where $\alpha$ = constant.

Show that the gauge transformation

$$A^\mu(x) \rightarrow A'^\mu(x) = A^\mu(x) + \partial^\mu \Lambda(x)$$

is also a symmetry of this Lagrangian